

Multi-Agent Market Simulation for Studying Strategy Behavior Under Transaction Costs and Market Regimes

1. Introduction

Transaction costs and market structure are first-order constraints on trading strategy profitability. While many strategy ideas are evaluated in frictionless or stationary environments, real markets impose fees, slippage, impact, and changing dynamics that can materially alter expected value (EV).

This work presents a multi-agent market simulation designed as a research sandbox for studying how strategy performance depends on microstructure and market-dynamics assumptions. The goal is not prediction or backtesting on real data, but controlled experimentation on incentives, expected value, and failure modes.

We study two core questions:

1. How do transaction costs affect profitability and risk?
2. How does performance depend on the underlying market environment?

2. Model

2.1 Market dynamics

The market is driven by a latent fair value F_t evolving according to parameterized dynamics. By tuning drift, mean-reversion strength, and noise, we construct environments that are:

- Trend-dominated
- Mean-reverting
- Noise-dominated (choppy)

Observed price:

$$P_t = F_t + \epsilon_t$$

Orders execute with:

- Per-unit fee
 - Per-unit slippage
 - Linear impact
 - Probabilistic fills
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2.2 Agents

- Momentum trader
- Mean-reversion trader
- Market maker
- Noise traders

All face position limits, leverage limits, and trading frictions.

2.3 Metrics

- Total PnL
- Sharpe-like ratio
- Turnover
- Max drawdown

All results averaged over 30 seeds.

3. Experimental Design

3.1 Experiment 1 — Transaction Costs

We sweep per-unit costs over:

{0,0.001,0.0025,0.005,0.01,0.02,0.05,0.1,0.2}{0,0.001,0.0025,0.005,0.01,0.02,0.05,0.1,0.2}

All other parameters fixed.

Hypothesis: high-turnover strategies degrade first.

3.2 Experiment 2 — Market-Dynamics Dominance

We construct:

- TREND world
- MEANREV world
- CHOP world

Friction set to zero to isolate environment effects.

Hypothesis: no universal winner.

4. Results

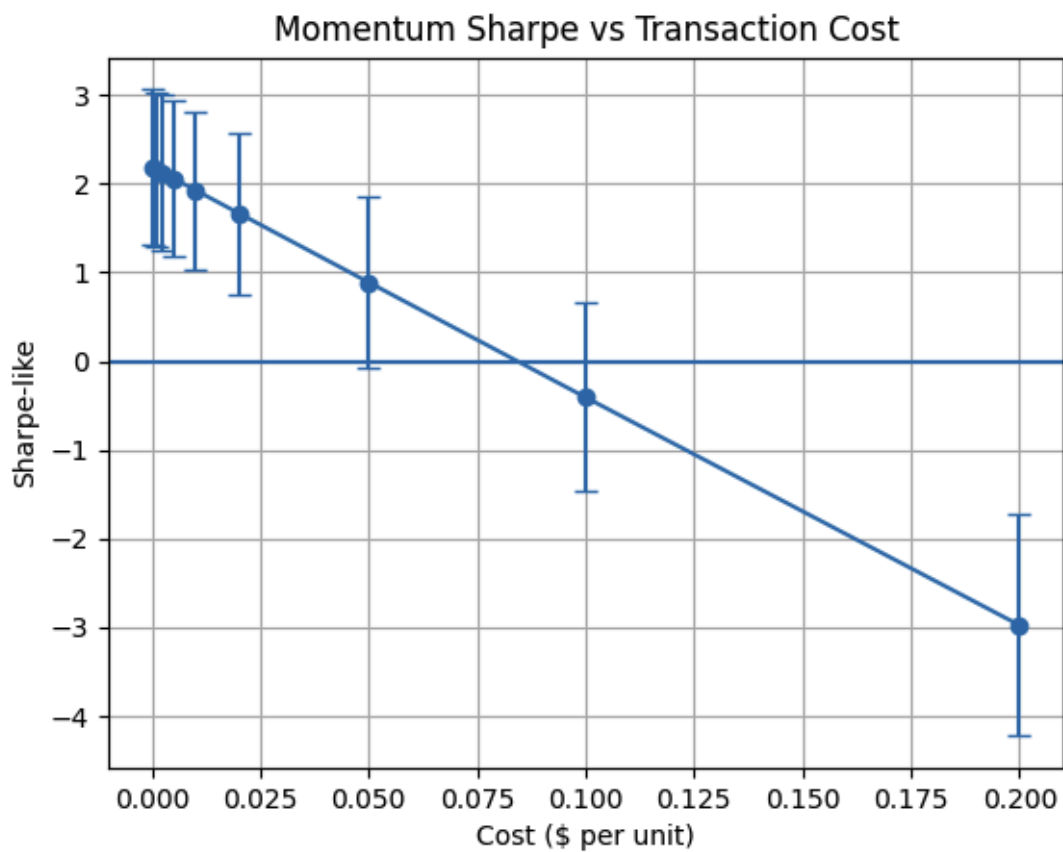
4.1 Transaction Costs (Experiment 1)

Momentum

Momentum performs strongly at low cost and decays smoothly as friction rises.

Sharpe crosses zero around cost ≈ 0.1 .

Figure 1 — Momentum Sharpe vs Transaction Cost

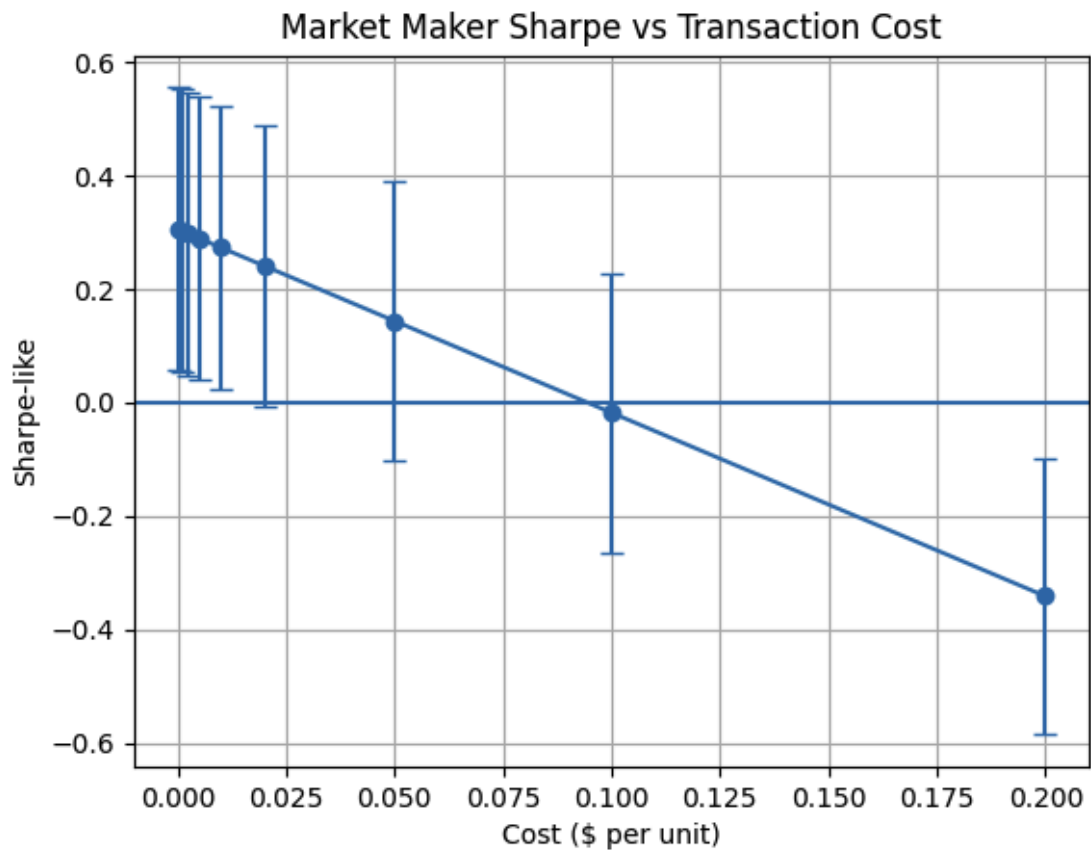


This illustrates a clear friction-driven phase transition.

Market Maker

The market maker starts with smaller but positive edge and decays more slowly.

Figure 2 — Market Maker Sharpe vs Transaction Cost



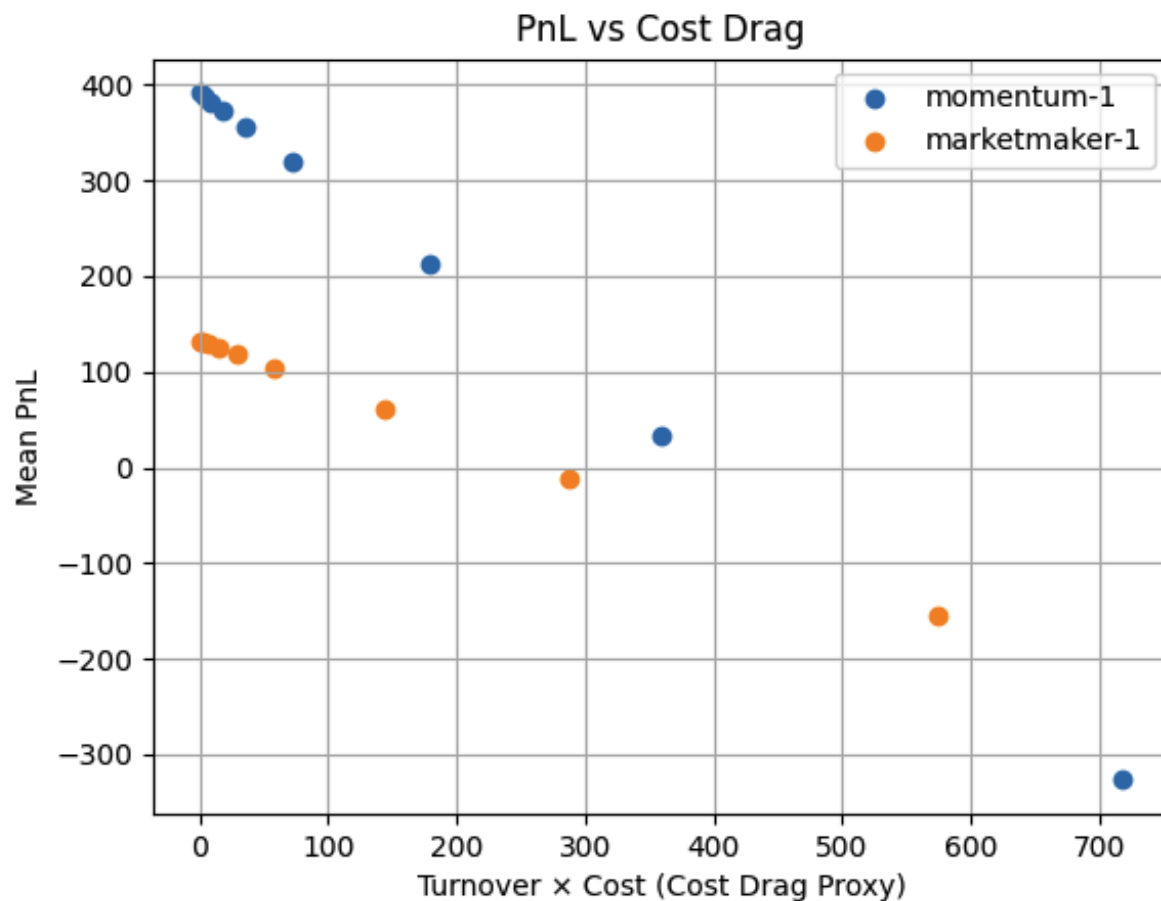
Spread capture is more robust than directional prediction but still cost-sensitive.

Mechanism: Turnover × Cost

Performance decay is well explained by:

$$\text{cost drag} \approx \text{turnover} \times \text{cost}$$

Figure 3 — Mean PnL vs Turnover × Cost



Profitability collapses once cost drag exceeds alpha.

Mean-reversion and noise

Mean-reversion is generally unprofitable.

Noise traders behave near zero at low cost and negative at high cost.

These serve as sanity checks

4.2 Market-Dynamics Dominance (Experiment 2)

Momentum shows extreme regime dependence:

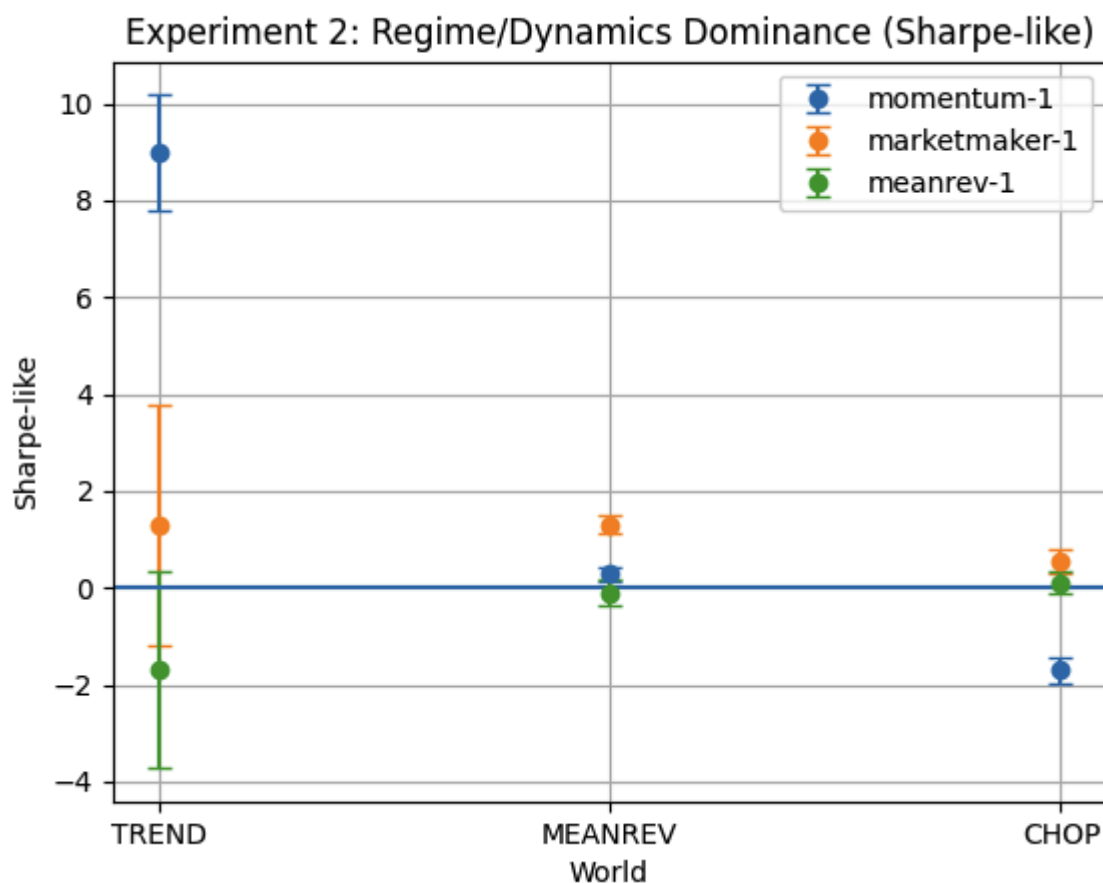
- Excellent in TREND
- Near zero in MEANREV
- Negative in CHOP

Market making remains positive across worlds.

Mean-reversion fails to exploit its intended environment.

Sharpe by world

Figure 4 — Sharpe-like by Market World

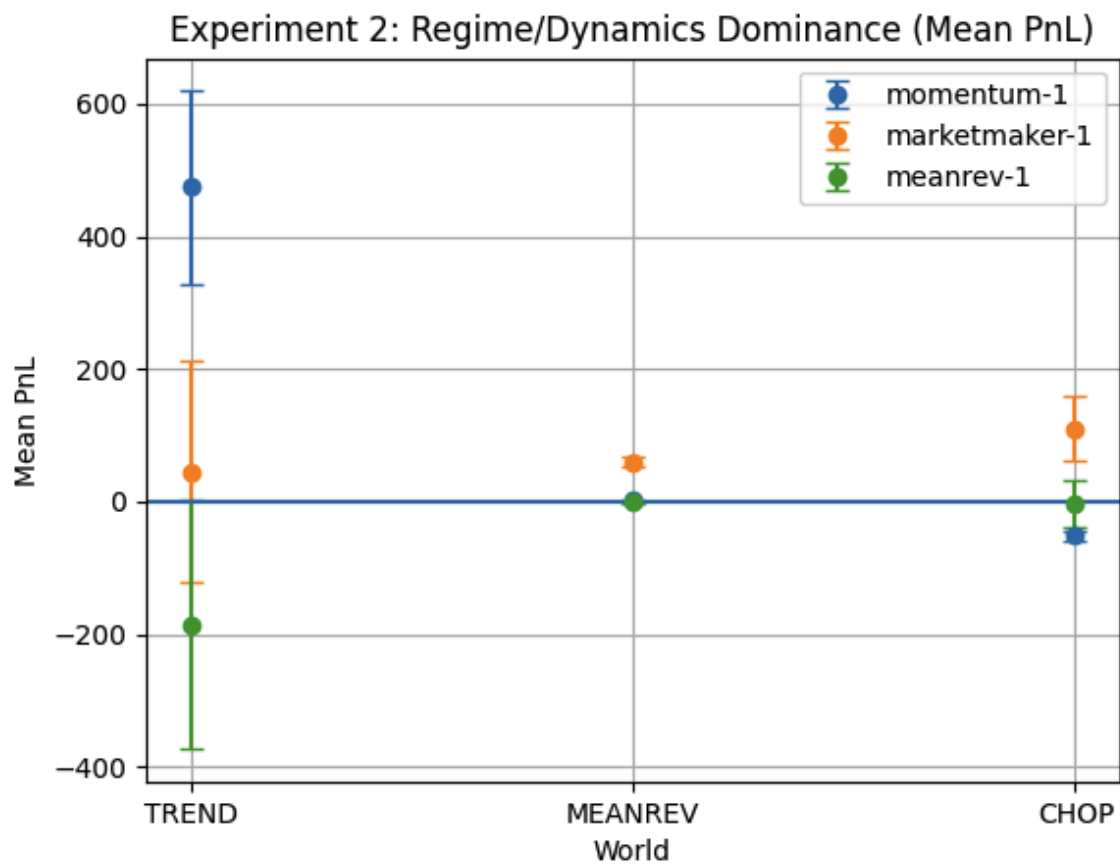


Momentum dominates in TREND but collapses in CHOP.

Market maker remains consistently positive.

PnL by world

Figure 5 — Mean PnL by Market World



This confirms that strategy success is environment-dependent.

5. Discussion

Several themes came out of these experiments that changed how I think about strategy evaluation.

First, alpha looks very conditional. A strategy's profitability isn't just about whether its signal is "good," but about the microstructure and environment it operates in. The same momentum logic that works extremely well in a trending world can quickly degrade or even become harmful in a noisy one. That reinforced for me that performance is always tied to assumptions about the market.

Second, high turnover turned out to be fragile. Turnover effectively scales your exposure to costs, so even a decent edge can get overwhelmed if a strategy trades too frequently. In the simulations, many of the performance differences under higher friction could be explained by the simple interaction between turnover and per-unit costs. That made the trade-off between activity and robustness very concrete.

Spread capture, on the other hand, looked comparatively robust. The market-making agent didn't dominate, but it stayed positive across more settings than the directional strategies. Intuitively, this makes sense: earning the spread is less dependent on correctly forecasting direction and more on providing liquidity. It highlighted to me why many real-world strategies focus on structural edges rather than pure prediction.

Another takeaway is that there really isn't a universal strategy. Different worlds favored different behaviors, and no agent was best everywhere. That was actually reassuring — it suggests the simulator isn't biased toward one style and that the results reflect genuine trade-offs.

Finally, the failure modes were as informative as the successes. The mean-reversion agent, for example, didn't perform well even in the mean-reverting environment. Rather than treating that as a problem, I see it as a useful signal: having the "right" idea isn't enough if the implementation, thresholds, or sizing are off. It points to where deeper design work is needed.

6. Limitations

This simulator is intentionally simplified, and I try to be careful about how I interpret the results.

It does not model latency races, adversarial agents, or a full limit order book. Those factors can matter a lot in real markets, especially for high-frequency or market-making strategies. The environment is also not calibrated to real data, so the magnitudes of Sharpe or PnL shouldn't be taken literally.

Because of that, I view the results as conceptual rather than predictive. The value of the framework is in understanding mechanisms — like how costs interact with turnover, or how strategy logic aligns with market structure — not in claiming that a specific strategy would work in production.

7. Conclusion

Two main findings:

1. Transaction costs induce a phase transition where alpha disappears.
2. Strategy performance is regime-dependent; no universal winner exists.

The dominant mechanisms are:

- Turnover-scaled cost drag
- Alignment between strategy and market structure

This framework is useful for reasoning about incentives and robustness under uncertainty.